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DISTRIBUTED OCEAN SENSING WITH WATER-LANDING UAVS AND CELLULAR COMMUNICATION

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ABSTRACT

This work presents the design and validation of a water sensing platform using compact, water-landing uncrewed aerial vehicles (UAVs) to perform measurements of water parameters, including salinity, temperature and speed. Each UAV had an onboard camera for selecting water-landing target regions, and a custom enclosure comprising onboard electronics for acquiring and processing water sensor data and communicating over cellular networks. Trajectories with multiple water-landings provided spatially-resolved measurements that were interpolated to create parameter maps. While landed on water, the UAV positions were tracked with Global Navigation Satellite System (GNSS) data to enable in situ measurements of surface water speed. A cloud software framework with bi-directional cellular communication to the UAVs ingested sensor data, provided a continuously-updated water parameter dashboard, and enabled control of sensing. Validation of the platform was performed on the Pacific Coast of Canada. This work represents a prominent step towards autonomous, swarm-based UAV monitoring of flood-prone regions that have been traditionally challenging and risk-intensive to access. This innovative and modular UAV-based platform is scalable to address a wide range of ocean sensing applications within the Blue Economy.

Keywords: UAV, Field Monitoring, Automation, Floating System, Resilience

1. INTRODUCTION

Frequent monitoring of coastal oceans and rivers is critical for preparedness and response to climate events. *In situ* water measurements are often required for monitoring parameters such as salinity and temperature that can be crucial for hydrological forecasting, informing ecosystem management, and developing effective long-term adaptation strategies.

Current *in situ* measurement methods have significant limitations that can manifest prominently during extreme weather events such as floods. For instance, water monitoring stations have fixed locations that preclude adaptive sampling strategies; floating sensorized buoys are not suited to targeting specific locations; manual water sampling and sensing are often resource intensive and can be impractical in many locations and during inclement weather.

This paper covers the design and development of a water-landing uncrewed aerial vehicle (UAV) system designed for collection and transmission of *in situ* salinity and temperature measurements. Sensor and Global Navigation Satellite System (GNSS) data were acquired concurrently with a custom module integrated onto the UAV. These data were transmitted over a cellular network and ingested into a cloud framework for storage, processing, and visualisation. While landed on the water surface, the UAV acted as a floater, thereby enabling measurements of local surface water speed and surface water path tracing.

2. RELATED WORKS

The use of UAVs for environmental monitoring has been rapidly increasing over the past two decades. Initial forays into the use of UAVs for large-scale environmental data collection demonstrated that they can provide an affordable alternative to manual sampling methods [1]. As multirotor UAV technology developed, UAVs were shown to be advantageous for precision water sampling, owing to their high maneuverability and low mechanical complexity [2]. Recent advancements in multirotor UAVs have facilitated advanced piloting and autonomous control [3]; however, the ability to efficiently acquire water data from UAVs in complex environments remains in the early stages of development.

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A common UAV-based water sampling paradigm involves attaching a water collection device below the UAV via a tether and hovering above the water while samples are collected [4][5][6] for subsequent laboratory analysis. Many early UAV-based water monitoring efforts were focused on measuring chlorophyll and turbidity; other parameters such as salinity have been reported less frequently [7]. Tether-based water sampling has been combined with *in situ* measurements, as shown by Dai et al. [8]. A significant limitation of these paradigms is that hovering while taking sensor measurements rapidly consumes UAV battery power, significantly limiting the number of distinct sensing locations during a single deployment. Specialized UAVs with water-landing capabilities have been shown to extend the operational time on water body surveys as they conserve battery power when resting on water [9][10]. Several schemes for integrating *in situ* sensing devices with water-landing UAVs have been considered, including systems for acquiring measurements at specific depths [11], hybrid UAVs that can travel on water surfaces, such as a hovercraft [12], and collaborative systems that pair *in situ* data collection with hyperspectral images [13].

Transmission of *in situ* water sensing data from UAVs can support agile decision making for end-users by improving hydrological monitoring efficiency [14]. Real-time transmission of *in situ* data from a tethered UAV was achieved using both cellular [8] and Long-Range Wide Area Network (LoRaWAN) communication [15]. In addition, cloud frameworks have been incorporated to provide low-latency data displays for environmental monitoring UAVs [16].

In addition to *in situ* sensor measurements, UAVs have also been used to gather data on surface water speed, a quantity that can be valuable in a wide range of applications, including flood assessments. As reviewed by Zhou et al. [17], these data can be acquired with non-contact and contact techniques. Tracking water dynamics from UAVs has primarily been studied using optical and Doppler radar techniques. The optimal use of these techniques depends significantly on environmental conditions. For instance, optical velocimetry techniques are dependent on the presence of visually discernible texture on the water surface or the use of seeding techniques in which floating particles are added to the water [18],[19]. Doppler radar techniques can be inaccurate in the presence of turbulent water and wind shearing on the water's surface [20][21]. During climate events with inclement weather and a high risk of drone loss, robust velocimetry techniques remain a significant challenge.

3. MATERIALS AND METHODS

Our system comprises three subsystems: a module that acquires and transmits sensor data, the water-landing drone to carry the module, and the cloud framework that stores, processes and displays data for remote visualization (Figure 1). This modularity was designed for flexibility; the subsystems can be interchanged to accommodate different system requirements that may arise.

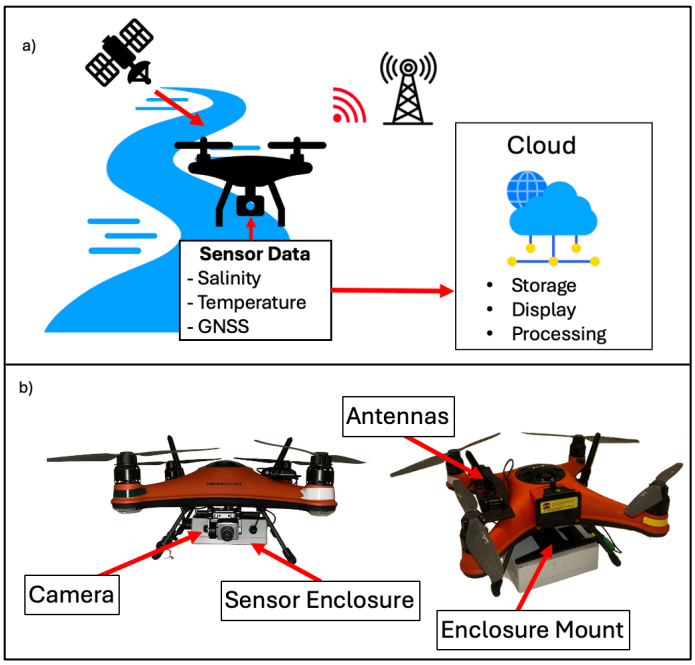


FIGURE 1: a) End-to-end system data flow. *In situ water* measurements are acquired by the UAV (Uncrewed Aerial Vehicle) in conjunction with Global Navigation Satellite System (GNSS) data; these sensor data are transmitted via cellular networks to the cloud. b) UAV with the custom enclosure that comprises the sensor and communications module, and the cellular and GNSS antennas.

Sensors within the module were integrated with a microcontroller (ESP32; Adafruit Feather) that interfaced with a cellular-enabled Internet of Things (IoT) communication module (Blues IoT Notecarrier F) with a waterproof antenna (Taoglass, Phoenix II) for data transmission to the cloud (Figure 2a).

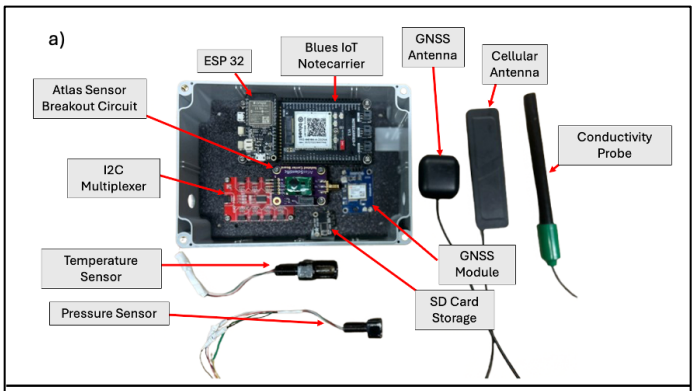
The onboard sensors included an electrode-based conductivity probe for salinity measurements (Atlas Scientific, Conductivity K 1.0; nominal range: 0.07 to 500,000 µS/cm) and a thermal sensor for water temperature (Blue Robotics, Celsius; nominal accuracy: 0.1 °C). The thermal sensor data were used for active temperature compensation of salinity measurements. A piezoresistive pressure sensor (Blue Robotics, Bar02; nominal accuracy: 0.14 mm in water, 13 cm in air) was used to estimate drone altitude and detect when the enclosure was submerged during drone water landings. The GNSS location data were acquired with a dedicated module (Widealiff, NEO-6M) and a waterproof antenna (Bingfu; SMA GPS Antenna). A solid state (SD) storage module (Adafruit, XTSD) was used for local data saving as a backup to cellular data transmission.

Each data point, in which sensor data were concatenated into JavaScript Object Notation (JSON) (Figure 2b), was transmitted by the IoT cellular communication module via a cloud-based IoT device management and data routing platform (Blues Notehub) to the cloud framework (Microsoft Azure). The microcontroller was programmed so that cellular communication was triggered upon water-landing, as detected by the pressure sensor. On

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battery power (1200 mAh; Lithium Polymer), the system could operate for 3 hours with continuous data transmission.

The module was housed in a waterproof (IP68) polymer electronics enclosure (Polycase), which was mounted to the drone via a custom 3D-printed adapter. Cables for the environmental sensors and the antennas (GNSS and cellular) passed through the waterproof enclosure using penetrators (Blue Robotics), sealed with high-grade epoxy (West System, G/Flex 650). The two antennas were positioned adjacent to the top surface of the drone using a second 3D-printed adapter. The components within the sensor and communications module were arranged for weight balancing, such that the centre of balance of the electronics module was directly in the middle of the enclosure. A small counterweight (50 g) was added to the UAV to counteract the residual inertial moment caused by a shift in centre of balance with the enclosure. It was found that centre of balance regulation is needed to ensure flight stability, particularly during takeoffs and landings. For the outdoor temperatures in which it was operated (< 25 ºC), neither active cooling nor specialised heat sink components were required for the electronics within the module.



b)


```
{
"event": "7a974d5a-0afb-4dbd-8010-2aebbc78a5d8",
"file": "data.qo",
"body": {
        "latitude": 49.321029,
        "longitude": 123. 140349167,
        "salinity": 2.91000,
        "temperature": 8.0823,
        "pressure": 1034.540039
        "enclosureTime": 2025-11-20T20:09:34.383813Z
        }
},
```


FIGURE 2: a) Electronic components of the sensor and cellular communications module. The enclosure is shown without penetrators and with the sensors and antennas disconnected. b) Representative example of a transmitted data point from a landing sequence on the Capilano River, where the "event" parameter provides a unique identifier for the data point, "file" indicates that the data point holds sensor data, "body" comprises acquired sensor and Global Navigation Satellite System (GNSS) data and a timestamp (UTC format).

The water-landing UAV (Swellpro Fisherman FD3) is waterproof (IP67) and can land and takeoff from water. This functionality allowed the drone to deliver the sensing module to multiple target locations consecutively. The payload capacity of the drone (nominal: 2 kg) was sufficient for the enclosure and adapter (combined: 0.69 kg). The drone was flown at distances less than 300 m from the operator (nominal maximum distance: 1.3 km). With the sensor and communications module, flight time was approximately 20 minutes.

Sensor calibration and testing was performed prior to field deployment. A three-point calibration of the salinity sensor was performed using reference solutions with known salinity (0 µS, 18,800 µS, and 80,000 µS). Across 10 measurements of other reference solutions with different salinity values that ranged from 8,237 µS to 56,364 µS, the calibrated sensor had a mean absolute relative error of 7.4%. The pre-calibrated temperature sensor had a nominal accuracy of 0.1°C; no further calibration was performed. When tested on land, the GNSS location accuracy had a mean error of 0.86 m; its minimum resolution was 0.42 m.

The dependencies of the cellular signal quality on the proximity of the antenna to water were tested with the module disconnected from the UAV. Four parameters provided by the IoT communication module were acquired: the reference signal strength indicator (RSSI), the reference signal received power (RSRP), the reference signal received quality (RSRQ), and the signal to interference plus noise ratio (SINR). These parameters directly relate to data handover success and throughput and are thus well suited to compare cellular signal conditions for an embedded system [22]. They were measured with the system on land, one meter above the water, and on water. The orientation of the antenna (horizontal or vertical) was also varied.

Preliminary validation of the system was performed on the Capilano River estuary in West Vancouver (Figure 3), with permissions obtained (Nav Canada and District of West Vancouver). The overall objective was to obtain an initial indication of operational feasibility. This deployment consisted of 11 landing sequences; each started with a water landing and terminated with a takeoff downstream. The landing positions were chosen manually with the use of the onboard UAV camera.

4. RESULTS

4.1 Cellular Signal Metrics

With the antenna positioned horizontally (the indicated orientation in the manufacturer's documentation), signal quality (RSRQ and SINR) was highest with the system placed over land (Table 1). Relative to this position, the quality was lower with placement 1 m above water and even lower when the system was resting on the water surface. The power of the reference signal (RSRP) was similar for the module on the ground and 1 m above the water, and lower when measured on the water surface. In contrast, the total received power (RSSI) was lowest on the ground and highest 1 m above water. Taken together, these

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results suggest that interference with reflections from the water surface likely contributed deleteriously to the signal quality.

With the antenna positioned vertically (90 degrees to its indicated orientation), there were different trends with the cellular signal metrics. In this orientation, all four metrics were suggestive of a potential advantage to vertically-oriented antennas for reception and transmission; however, owing to the complexity cellular signal propagation and the diverse water environments that can be encountered, further testing is required to draw more definitive conclusions.

TABLE 1: Cellular signal metrics

Antenna Orientation	Scenario	RSRQ (dB)	SINR	RSRP (dB)	RSSI (dB)
Horizontal	On land	-11.05	4.35	-81.6	-54
	1 m above water	-12.6	3.75	-81.5	-49.95
	On water	-13.55	2.35	-85.2	-52.15
Vertical	On land	-12.8	2.8	-86.3	-55.75
	1 m above water	-12	5.55	-77.15	-45.75
	On water	-12.55	1.25	-84.4	-53.7

Reference signal received quality (RSRQ), signal to noise ratio (SINR) reference signal received power (RSRP), reference signal strength indicator (RSSI), as provided by the cellular communications module, varied with distance of the drone to the water surface, and for different antenna orientations.

4.2 Cloud Data Streaming

During water-landing sequences, cellular connectivity was maintained; all data that were saved locally were also received by the cloud framework. In total, 1007 data points were sent, comprising sensor measurements, GNSS data, and a timestamp. Over the study, the average data rate was 25.8 data points per minute. However, the duration between transmitted data points varied. For each data point, the average transmission latency, defined as the time interval between start of transmission and completion of data reception on the cloud framework for a single data point, was 4.22 seconds. Of the 1007 data points transmitted, there were 5 cases where this latency was more than 10 seconds. The longest latency for a data point was 43.8 seconds; during this time, no data were sent. These latencies could have arisen from a multitude of delays during transmission over the cellular network and propagation through cloud intermediaries.

4.3 Data Plots

Sensor and GNSS data acquired from the Capilano River deployment (Figure 3) were processed with Geographic Information System (GIS) software (QGIS, QGIS Foundation) to obtain salinity, temperature and water speed maps. On all of the gradient maps (Figures 4, 5, 6), each of the drone landing locations are marked and labeled chronologically. Interpolation with inverse distance weighting (IDW) was used to estimate measurement values at arbitrary positions $\boldsymbol{x}$ on the map from measured locations $\boldsymbol{x_i}$. Interpolated values $\boldsymbol{u(x)}$ were obtained using the inverse of their distance to the measured points, $\boldsymbol{d(x, x_i)}$.

$$u(x) = \begin{cases} \frac{\sum_{i=1}^{N} w_i(x) u_i}{\sum_{i=1}^{N} w_i} \\ u_i \end{cases}, \qquad (1)$$

in which the weights, $\boldsymbol{w_i}$ were calculated as the inverse of the distance, $\boldsymbol{d}$, between points to a power $\boldsymbol{p}$ (chosen as 3 in this study):

$$w_i = \frac{1}{d(x,x_i)^p}. \qquad (2)$$



FIGURE 3: Uncrewed Aerial Vehicle (UAV) system comprising a sensorized water-landing drone travelling between target points during deployment in the Capilano River estuary (West Vancouver, Canada).

The gradient map for *in situ* salinity measurements showed an overall increase from locations in the freshwater river towards the ocean (Figure 4a), as expected. However, the final two landing sequences that corresponded to a return of the UAV to upstream locations were incongruent with this trend, perhaps due to a tidal increase. The variation of salinity data within individual sequences was lower upstream in the river (Figure 4b), as compared to downstream (Figure 4c), which may have originated from the complex dynamics of river and ocean water mixing in the estuary.

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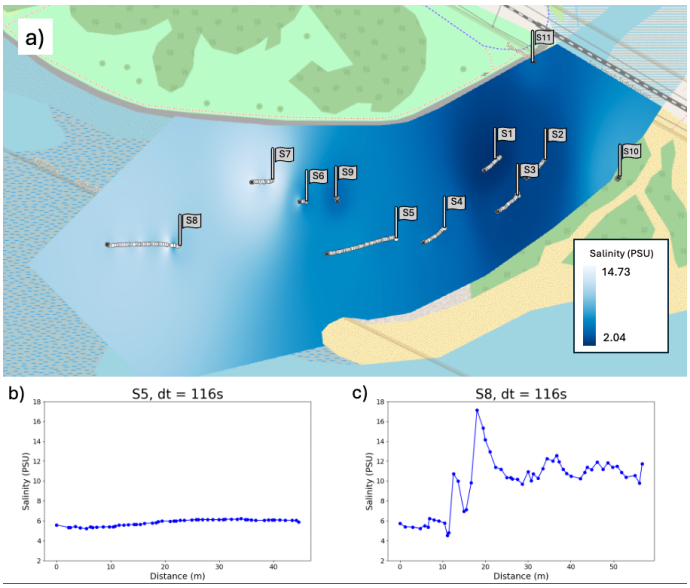


FIGURE 4: a) Salinity parameter map acquired from the Capilano River estuary; b) salinity data from landing sequence 5; c) salinity data from landing sequence 8. Each data point is marked on spatial parameter maps with a grey dot. The Uncrewed Aerial Vehicle (UAV) landing locations are marked with flags labeled chronologically S1 to S11. The UAV takeoff locations are marked with a dark dot. In landing sequence 5, the standard deviation from the mean was 5.36%, which contrasts with 27.87% in landing sequence 8.

The *in situ* temperature parameter map had a similar appearance to the salinity gradient map: the measured temperature increased with decreasing distance to the ocean, and there were localised temperature spikes in the last two sequences (Figure 5a). Temperature variations within sequences were lower upstream (Figure 5b) than downstream (Figure 5c).

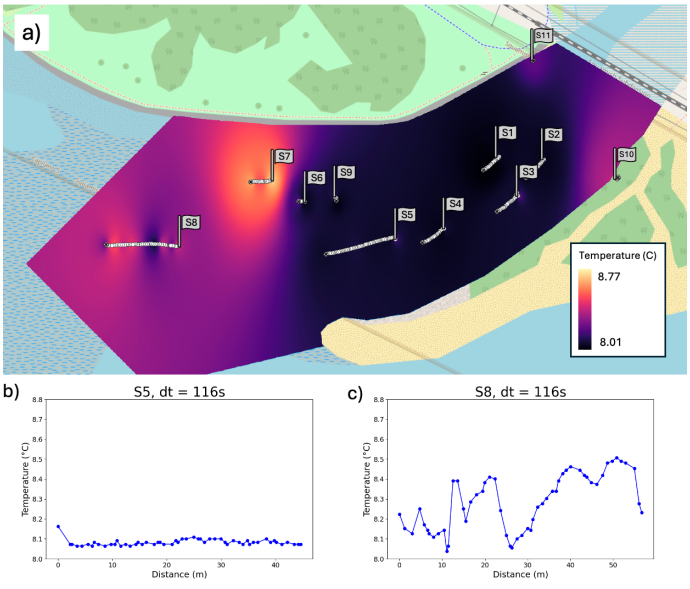


FIGURE 5: a) Capilano River estuary temperature gradient, b) temperature data from landing sequence 5, c) temperature data from landing sequence 8. The data are displayed in the same manner as in Figure 4. In landing sequence 5, the standard deviation from the mean was 0.14%, which contrasts with 1.70% in landing sequence 8.

In situ surface water speed was estimated using the GNSS locations and timestamps, with a 3-point moving average applied to mitigate location errors. Overall, the surface water speed increased with distance to the ocean (Figure 6a). The origin of this increase is unknown and is likely multifactorial. Unlike salinity and temperature measurements, variations in measured speed within sequences were similar across locations (Figure 6b, c); for example, standard deviation from the mean was 17% in landing sequence 5 and 17% in landing sequence 8.

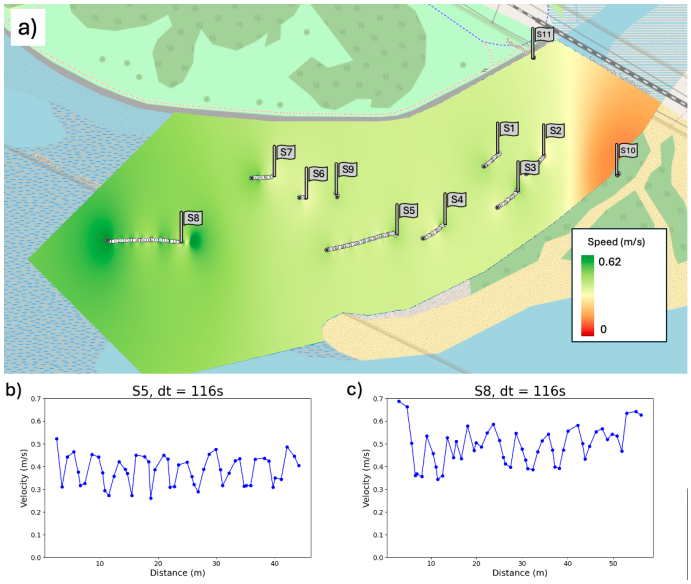


FIGURE 6: a) Capilano River estuary speed gradient; b) speed data from landing sequence 5, c) speed data from landing sequence 8. The data are displayed in the same manner as in Figure 4.

5. DISCUSSION AND CONCLUSIONS

This study provided a preliminary validation of a system comprising a water-landing UAV that acquires distributed *in situ* salinity, temperature and water speed measurements with concurrent communication with a cloud framework intended for dynamic riverine and coastal environments prone to flooding.

A key innovation of this study was the integration of cellular data transmission with a water-landing UAV platform for *in situ* water salinity, temperature and flow data for readily-accessible water measurements. This capability provides an important step forward in reducing the latency of distributing actionable data during acute climate events such as floods. Although the signal quality tests indicated that the proximity of the horizontally-positioned cellular antenna to water was deleterious to transmission, communication between the UAV and the cloud

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remained reliable during deployment in the Capilano River estuary. In future implementations of this UAV system, cellular communication will offer the potential for low-latency cloud-based UAV control, for instance with the use of 5G or 6G. Alternatives such as satellite transmission are compelling for broader geographic coverage, relative to cellular transmission; however, this advantage may have the corresponding disadvantage of lower transmission rates.

To the authors' knowledge, the use of a floating UAV with GNSS measurements to obtain water speed measurements is novel. Previous studies have validated the use of GNSS-based velocity measurements from floating buoys, demonstrating their ability to capture spatial variability and flow dynamics [23], [24]. These studies show that GNSS drifters can provide robust proxy measurements of water surface velocity, with accuracy generally improving at higher flow speeds and with repeated deployments. Extending this approach for UAV deployment adds significant value by allowing for repeated measurements cycles without the need for physical retrievals of the device, as the platform can return to a starting location using its flight capabilities.

Established UAV-based methods for water velocity measurements such as vision-based velocimetry and Doppler radar have been shown to have high accuracy during favourable weather conditions [17]. For instance, UAV-mounted Doppler radar measurements had errors between 1-3% under favourable hydrological conditions as compared to conventional measurement techniques; however, errors increasing to ca. 20% in more complex flow and lighting conditions [19]. Similarly, video-based surface velocity methods validated by Sy et al. reported surface velocity with errors below 5% relative to Acoustic Doppler Current profiler (ADCP) measurements [18]. While, these methods set high benchmarks for accuracy in ideal river conditions, GNSS-based measurements from a floating UAV may offer improved robustness during turbulent or adverse weather conditions, as they do not rely on visually discernable surface patterns or acoustic reverberations, both of which can be strongly affected by wind, rain and surface disturbances. Floating GNSS measurements are prone to wind-induced drift that could introduce bias into flow estimates.

With respect to system robustness and cost, a floating-UAV collecting GNSS measurements may offer additional advantages over alternative UAV-based approaches. Acquiring water speed data without the need for auxiliary sensors such Doppler radar can reduce the overall system cost and minimize the effect of heavy, wind-catching payloads on flight stability during inclement weather conditions. Moreover, acquiring ocean sensor data while landed conserves UAV battery as compared with hovering above the water, allowing for longer data collection periods. Concurrent acquisition of *in situ* water parameters such as salinity and temperature provides valuable contextual information in coastal flooding where heavy rainfall and freshwater discharge play critical role. Overall, the relative advantages of a floating-UAV system depend on the environmental conditions and the monitoring objectives for which the system is deployed.

The presented UAV system is situated within a broader ecosystem of river monitoring techniques including buoys for *in situ* quality measurements, uncrewed surface vehicles (USV), underwater remotely operated vehicles (ROV) and satellite-based monitoring. Each of these platforms has their advantages; for instance, buoy systems are robust and have low mechanical complexity; USVs are subject to fewer regulations than UAVs; underwater ROVs are well fit for capturing information far below the water's surface; satellite monitoring provides a consistent source of overhead visual data conducive to river boundary monitoring and optical properties. However, UAVs provide unique benefits as they amenable to rapid deployment, rapidly gathering data from a wide range of locations with concurrent visual imagery. Several UAV systems presented in the literature enabled measurement of water parameters by lowering a tethered sensor module into the water rather than landing [4,5,6,8]. While these approaches reduce the risk of vehicle loss associated with water landings, a water-landing drone that can obtain sensor data whilst shutting off power to the motors can acquire data over longer timescales.

Recent advances in edge computing and cloud communication can be expected to greatly decrease barriers to deploying sensorized UAVs in complex environments such as those that arise during flooding. These developments could increase accessibility among non-specialists, including members of coastal communities with extensive insights about their water who could assist with scaling-up deployments. Swarms of UAVs, connected through IoT frameworks with cellular communication and with corresponding cloud-based data displays and AI processing, could significantly transform the efficiency of responses to climate events.

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